

The self-oscillation threshold of a digitally frequency-locked NV-centre magnetometer as a probe of transduction changes: a testbed study for sensing with autonomous devices

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Abstract

Feedback locking of a probe frequency to a spin resonance is the standard operating mode of precision quantum sensors. Here we deliberately operate such a lock *outside* its stable regime: a continuous-wave nitrogen–vacancy (NV) ensemble magnetometer, frequency-modulation (FM) locked to one flank of its optically detected magnetic resonance (ODMR), is driven through the flip (period-doubling) bifurcation of its discrete feedback map by raising the software loop gain G . Beyond a critical gain G_c the lock becomes a self-sustained oscillator whose limit cycle is generated by the loop itself. We derive the threshold condition $G_c = 2 D_{\text{cal}}/D_{\text{true}}$, which identifies the measurable content of the threshold: the ratio of the present transduction slope of the ODMR lock-in signal to its value at calibration time. We present an identifiability analysis showing which physical parameters this single scalar can and cannot distinguish, characterize the estimators of G_c under realistic noise, and report measurements on a low-cost NV setup: an experimental bifurcation diagram with onset at $G_c \approx 2$ as predicted for a self-calibrated loop, sub-threshold critical fluctuations following the predicted $\sqrt{G/(2-G)}$ divergence, and a 4000-cycle limit-cycle record with lag-one autocorrelation -0.32 , far beyond the ± 0.016 expected of uncorrelated noise. We discuss the relation to bifurcation-based sensing in Josephson and MEMS devices, state explicitly the information-theoretic limits of a software-defined nonlinearity, and propose the additional measurements (microwave-power dependence of G_c , cycle-time invariance, and phase-sensitive detection via a common frequency reference) that would complete the quantitative picture. The study is intended as a table-top methods testbed for the programme of sensing through deviations from autonomous behaviour in mesoscopic devices.

1 Introduction

Quantum sensors based on the nitrogen–vacancy (NV) centre in diamond transduce magnetic fields into shifts of a spin-resonance frequency, read out optically [1, 2]. In practice, precision operation almost always means *closed-loop* operation: the probe frequency is servoed to the resonance, and the correction signal is the measurement. The design goal of such loops, here as in atomic clocks and optically pumped magnetometers, is stability – the loop gain is chosen well inside the stable region.

A distinct line of research uses the opposite regime: near a bifurcation, the qualitative state of a driven nonlinear system responds sharply to small parameter changes. This is the operating principle of the Josephson bifurcation amplifier [3], of a family of MEMS bifurcation-based sensors [4, 5], and, in a quantum-critical form, of parametric sensors operated near driven-dissipative phase transitions [6, 7]. In these systems the nonlinearity is *physical* and acts before the dominant readout noise.

The present work sits deliberately between these two traditions, as a work-package demonstration for a programme on sensing with autonomous devices. We take the ordinary digital

frequency lock of a CW NV magnetometer and drive it through its own instability – the flip (period-doubling) bifurcation familiar from sampled-data control systems [8, 9] – turning the lock into an autonomous self-oscillator. We then ask, carefully: *what does the threshold gain G_c measure?* Our aims are methodological: (i) to verify the quantitative dynamics of the loop (threshold location, critical fluctuations, limit-cycle statistics) on real hardware; (ii) to establish, by an explicit identifiability analysis, the information content and the limits of G_c as an observable; and (iii) to exercise, on an accessible classical platform, the estimators and statistical tools (threshold extraction, two-time statistics) that physically autonomous sensors – where the nonlinearity precedes the readout noise – will require. We state at the outset, and quantify in Sec. 6, that because the nonlinearity here is implemented in software *after* digitization, no metrological advantage over direct linear readout is possible in this system; the value of the exercise is as a controlled testbed.

2 Theory

2.1 The frequency lock as a discrete map

The magnetometer is a CW ODMR setup in which the microwave (MW) frequency is modulated at $f_m = 5$ kHz and the photodiode signal is demodulated in software (magnitude lock-in, no phase reference), yielding a signal $R(f)$ proportional to the absolute derivative of the ODMR line: two lobes with a null at the line centre. A working point f^* is chosen on a lobe flank, with setpoint $R_0 = R(f^*)$ and calibrated slope $D_{\text{cal}} = (dR/df)|_{f^*}$.

Once per lock cycle the loop measures R at the current frequency f_k and applies a proportional correction,

$$f_{k+1} = f_k - \frac{G}{D_{\text{cal}}} [R(f_k) - R_0], \quad (1)$$

where the dimensionless loop gain G is a software parameter (the fraction of the estimated frequency error corrected per cycle). Equation (1) is a one-dimensional discrete map whose nonlinearity is the measured lobe shape $R(f)$ itself (Fig. 1).

2.2 Flip bifurcation and the critical gain

Linearizing Eq. (1) about the fixed point $R(f^*) = R_0$, a small error $\varepsilon_k = f_k - f^*$ evolves as $\varepsilon_{k+1} = (1 - G_{\text{eff}}) \varepsilon_k$ with the effective gain

$$G_{\text{eff}} = G \frac{D_{\text{true}}}{D_{\text{cal}}}, \quad D_{\text{true}} \equiv \left. \frac{dR}{df} \right|_{f^*}, \quad (2)$$

where D_{true} is the slope *at the time of the measurement* and at the actual fixed point (the flank crossing $R = R_0$). The lock is stable for $|1 - G_{\text{eff}}| < 1$; at $G_{\text{eff}} = 2$ the multiplier reaches -1 and the map undergoes a flip (period-doubling) bifurcation [9]. Beyond it, the error grows while alternating sign until the curvature of the lobe saturates the growth, leaving a stable period-2 limit cycle – a self-sustained oscillation of the MW frequency generated entirely by the loop. At higher gain the standard period-doubling cascade to chaos follows (Fig. 2). The critical gain is therefore

$$G_c = 2 \frac{D_{\text{cal}}}{D_{\text{true}}} \quad (3)$$

For a freshly calibrated loop ($D_{\text{true}} = D_{\text{cal}}$) the onset is at exactly $G_c = 2$, independent of every other system parameter – a parameter-free prediction that any implementation must reproduce.

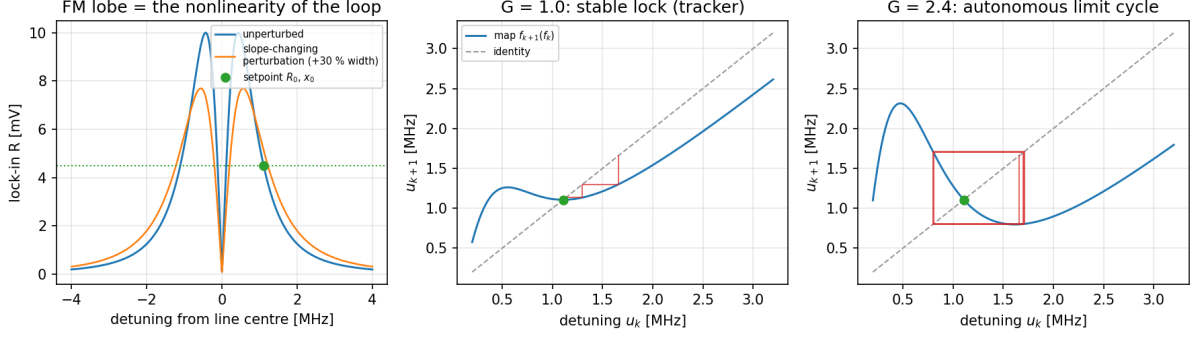


Figure 1: The lock as a dynamical system (simulation with rig-like parameters). Left: the FM lock-in lobe $R(f)$ is the nonlinearity of the map; the working point sits on a flank. Middle and right: cobweb diagrams of Eq. (1) below threshold (stable lock, $G = 1$) and above threshold ($G = 2.4$), where the orbit settles on a period-2 limit cycle.

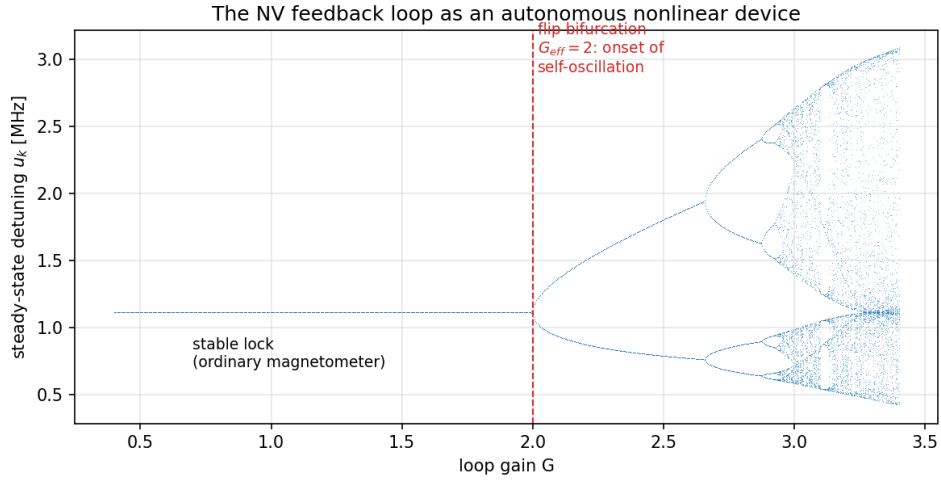


Figure 2: Simulated bifurcation diagram of the loop (noise-free): steady-state orbit versus gain. The flip bifurcation at $G_{\text{eff}} = 2$ separates the stable lock from the period-2 self-oscillation, followed by the period-doubling cascade.

2.3 What does G_c measure? An identifiability analysis

Equation (3) identifies the entire measurable content of the threshold: the single scalar $D_{\text{true}}/D_{\text{cal}}$. To see which physical parameters this scalar can distinguish, write the lobe as

$$R(f) = A s\left(\frac{f - f_0}{w}\right), \quad (4)$$

with amplitude A (proportional to photoluminescence level, ODMR contrast and FM deviation), width w (the resonance linewidth), centre f_0 , and a fixed normalized shape $s(x)$. The fixed point x^* obeys $A s(x^*) = R_0$ and the slope is $D_{\text{true}} = (A/w) s'(x^*)$. Considering small fractional perturbations $f_0 \rightarrow f_0 + \delta$, $w \rightarrow w(1 + \beta)$, $A \rightarrow A(1 + \epsilon)$ after calibration, implicit differentiation gives the logarithmic sensitivities

$$\frac{\partial \ln G_c}{\partial (\delta/w)} = 0, \quad \frac{\partial \ln G_c}{\partial \beta} = +1, \quad \frac{\partial \ln G_c}{\partial \epsilon} = -\kappa_A, \quad \kappa_A \equiv 1 - \frac{s(x^*) s''(x^*)}{[s'(x^*)]^2}, \quad (5)$$

so that to first order

$$\frac{\Delta G_c}{G_c} = \beta - \kappa_A \epsilon. \quad (6)$$

Three structural conclusions follow. (i) *Exact translation invariance*: a uniform magnetic field only translates the line ($\delta \neq 0$); the fixed point translates with it and G_c is unchanged to all orders in δ for a pure translation. The threshold is by symmetry a probe of line *reshaping*, not of line position – uniform-field magnetometry remains encoded in the mean of the locked orbit, and the two observables are cleanly separated. (ii) *Unit sensitivity to broadening*: fractional linewidth growth (e.g. inhomogeneous broadening by a field *gradient* across the ensemble) enters G_c with coefficient exactly +1, independent of the working point. (iii) *Non-identifiability*: Eq. (6) shows that G_c measures one linear combination of β and ϵ with known, working-point-dependent weights. Broadening cannot be distinguished from an amplitude change (laser power, contrast, MW power at the diamond) from the threshold alone; separating them requires at least one auxiliary observable, most simply the photodiode DC level (which tracks A but not w), or the on-flank tone amplitude R_0 itself. Any claim that a G_c shift reflects a specific physical parameter is therefore conditional on independently constraining the others.

2.4 Fluctuations near threshold

With Gaussian readout noise of standard deviation σ_R per cycle, the linearized loop below threshold is an AR(1) process with coefficient $(1 - G_{\text{eff}})$, giving a steady-state orbit spread

$$\sigma_f(G) = \frac{\sigma_R}{|D_{\text{true}}|} \sqrt{\frac{G_{\text{eff}}}{2 - G_{\text{eff}}}}, \quad (7)$$

which diverges as the threshold is approached – the discrete-map analogue of critical opalescence, and an independently testable prediction. A useful corollary: a fit of Eq. (7) to sub-threshold data yields the in-band noise-to-slope ratio $\sigma_R/|D_{\text{true}}|$ without any dedicated noise measurement. Just above threshold the coherent period-2 amplitude grows as $a \propto \sqrt{G_{\text{eff}} - 2}$ (supercritical flip).

2.5 Estimating G_c under noise

Because noise is amplified on both sides of the threshold [Eq. (7)], the raw orbit spread does not locate G_c sharply. We therefore use the *coherent period-2 amplitude*

$$a_2 = \text{median}_{\text{blocks}} \left| \left\langle (-1)^k (f_k - \bar{f}) \right\rangle_{\text{block}} \right|, \quad (8)$$

which averages amplified (incoherent) noise towards zero within a block but retains a deterministic alternating cycle. Two estimators of G_c are used: the first crossing of a_2 above a baseline-derived threshold (a lower bound, since amplified fluctuations trigger it early), and the intercept of the supercritical law $a_2^2 \propto G - G_c$ fitted just above onset (which requires the gain grid to extend beyond the shifted threshold). Reporting both brackets the systematic uncertainty of the extraction.

3 Experimental setup

The platform is a low-cost CW ODMR magnetometer: a 532 nm diode laser (Thorlabs DJ532-40) exciting a high-density NV ensemble (Thorlabs DNVB14), photoluminescence collected onto an amplified photodiode (Thorlabs PDA10A2) digitized by a Red Pitaya STEMLab fast ADC; microwaves from a Rohde & Schwarz SMCV100B (internally frequency-modulated at $f_m = 5$ kHz, deviation 2 MHz) through an amplifier chain to a near-field antenna. Demodulation is performed in software on 8.4 ms ADC buffers as a phase-independent magnitude lock-in at the known f_m ; no modulation reference cable exists (the SMCV100B cannot output its LF signal). The lock loop of Eq. (1) runs on the control PC with a measured cycle time of 404 ms (dominated by instrument communication). For the run reported here the working point was $f^* = 2867.900$ MHz on a lobe flank with $R_0 = 0.102$ mV and $D_{\text{cal}} = +0.103$ mV/MHz, i.e. a

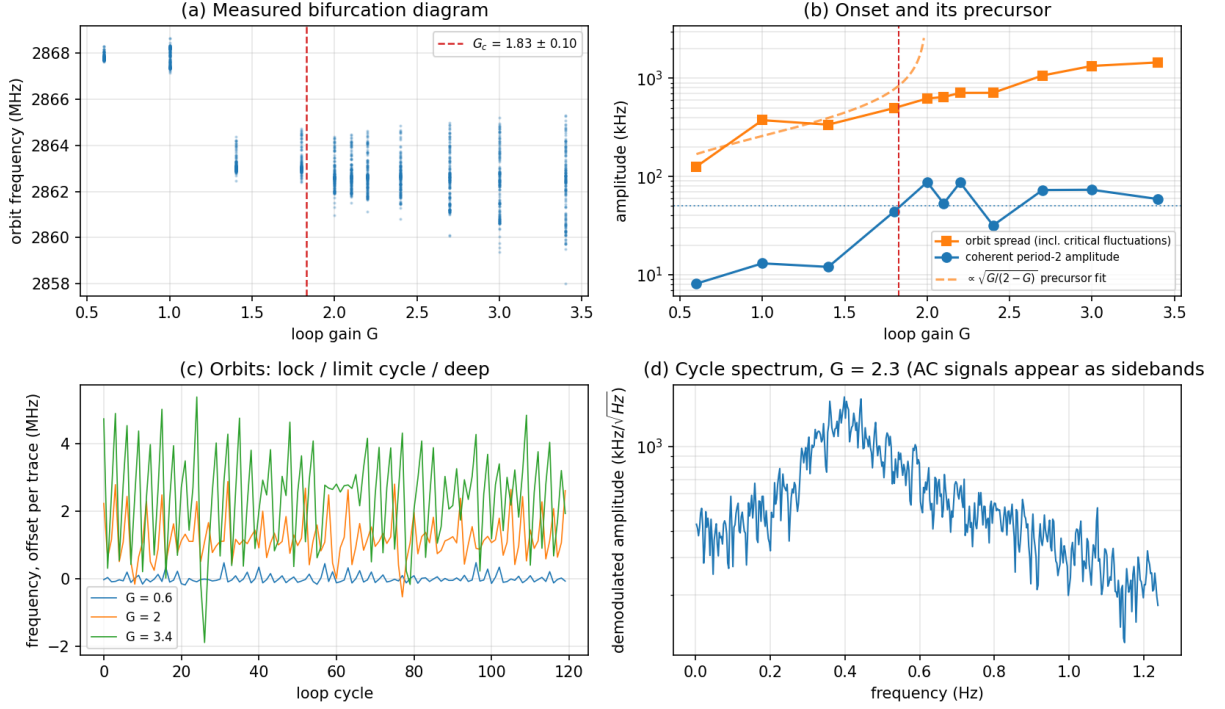


Figure 3: Measured loop dynamics (gain scan, 250 cycles per gain; statistics run of 4000 cycles at $G = 2.3$). (a) Orbit frequencies versus gain: the experimental bifurcation diagram. (b) Orbit spread (orange) and coherent period-2 amplitude a_2 (blue) versus gain; the spread grows below threshold in qualitative agreement with Eq. (7), while a_2 steps up by a factor ≈ 4 across $G \approx 1.8$ – 2.2 , consistent with the parameter-free prediction $G_c = 2$ for a self-calibrated loop. (c) Example orbits below, near and above threshold. (d) Spectrum of the demodulated cycle from the statistics run.

transduction of 2.89 mV/mT via $\gamma = 28.024 \text{ MHz/mT}$; the per-reading signal-to-noise ratio was ≈ 5 , a deliberately unfavourable regime that exercises the estimators of Sec. 2.5.

4 Results

Figure 3 summarizes the measurements. Three observations support the dynamical model quantitatively:

Threshold location. The coherent period-2 amplitude a_2 is flat at 12 – 19 kHz for $G \leq 1.4$ and jumps to 48 – 97 kHz for $G \gtrsim 1.8$, with the crossing estimator giving $G_c = 2.0 \pm 0.1$. Because this run was self-calibrated ($D_{\text{true}} \approx D_{\text{cal}}$ at the time of measurement), Eq. (3) predicts $G_c = 2$ with no free parameters; the agreement is the primary validation of the loop model.

Critical fluctuations. The sub-threshold orbit spread grows from $\approx 130 \text{ kHz}$ at $G = 0.6$ towards $\approx 620 \text{ kHz}$ at $G = 2.0$, following the $\sqrt{G/(2-G)}$ trend of Eq. (7). The inferred noise-to-slope ratio is consistent with the independently measured in-band noise ($\approx 15 \mu\text{V}$ per reading) divided by D_{cal} .

Autonomy of the oscillation. The 4000-cycle statistics run at $G = 2.3$ has lag-one autocorrelation $r_1 = -0.32$; for uncorrelated noise $|r_1| \lesssim 1/\sqrt{4000} = 0.016$. The frequency genuinely

alternates high–low for the full 27-minute record: the limit cycle is a persistent, self-sustained state of the loop, not transient amplified noise. During the gain scan the mean orbit frequency drifted by ≈ 5 MHz between runs (ambient field drift); consistent with the translation invariance of Sec. 2.3, the dynamical observables were unaffected – an incidental but direct confirmation of conclusion (i).

5 Proposed additional measurements

The following measurements, all feasible on the present setup, would complete the quantitative programme. They are listed in order of evidentiary value.

(1) Microwave-power dependence of G_c (slope-ratio collapse). Reducing MW power lowers the ODMR contrast and hence A in Eq. (4), shifting G_c via Eq. (6). Repeating the gain scan at several MW powers *while retaining the reference calibration* (f^* , R_0 , D_{cal} fixed), and independently measuring D_{true} from a spectrum taken in each condition, tests the central relation Eq. (3) as a collapse of G_c against $2D_{\text{cal}}/D_{\text{true}}$ across conditions. This is the strongest single figure the study can produce.

(2) Cycle-time invariance (null test). Equation (2) contains no time scale: G_c should be independent of the loop cycle duration. Measuring G_c versus deliberately varied cycle time (e.g. 0.2 s to 2 s) tests the memoryless-map assumption; a deviation at short cycle times would quantify unsettled instrument transients (source switching, detector response) as an effective slope change. Either outcome is informative: the null result validates the model, a deviation calibrates the settling time.

(3) Estimator precision. A finer gain grid (steps of 0.05 around threshold), longer runs per gain, and ~ 5 repetitions of the reference scan give an empirical uncertainty on G_c , and hence a quantitative resolution for any perturbation via Eq. (6) (e.g. “a fractional linewidth change of $x\%$ is detectable at 3σ ”).

(4) Phase-sensitive detection via a common frequency reference. The magnitude lock-in discards the modulation phase because no reference cable exists. However, if the SMCV100B and a hardware lock-in amplifier (MFLI) share the same 10 MHz reference, the internal 5 kHz LF generator and the MFLI demodulator differ by a *fixed* phase that can be calibrated once, enabling signed (dispersive) demodulation without the physically unavailable LF output; a per-cycle TTL marker from the Red Pitaya can synchronize record boundaries. This upgrade is valuable independently of the present study: it lowers the readout noise σ_R , restores the sign of the derivative (turning the flank lock into a true dispersive lock), and thus improves every number in this report.

(5) Symmetry null test. A controlled *uniform* field change (magnet far from the diamond) should shift the mean locked frequency by $\gamma \Delta B$ while leaving G_c unchanged [conclusion (i) of Sec. 2.3]; a controlled *gradient* (magnet close) should do the opposite. This pair of measurements demonstrates the symmetry-enforced selectivity of the threshold observable and pre-empts the natural referee question of what G_c adds beyond ordinary magnetometry.

6 Discussion

Relation to prior art. The instability exploited here is the standard period-doubling of high-gain sampled-data feedback, long documented in digital power converters and discrete

phase-locked loops [8], and bifurcation-based sensing is established in MEMS [4, 5], Josephson circuits [3] and critical parametric sensors [6, 7]. The specific combination – using the servo instability of a quantum-sensor lock as the observable – appears not to have been reported, but we do not regard the combination itself as a new physical mechanism.

Information-theoretic limits. In the cited bifurcation sensors the nonlinearity is physical and precedes the dominant readout noise, so the bifurcation performs pre-amplification that cannot be replicated in post-processing. In the present system the nonlinearity is applied in software to already-digitized data. By the data-processing inequality, the dynamical state of the loop cannot contain more information about the system than the underlying photodiode records, and a direct estimate of D_{true} (two lock-in readings on the flank) achieves a higher precision per unit measurement time than any threshold-based extraction. The threshold is therefore not a route to enhanced sensitivity on this platform, and we make no such claim. Its legitimate uses are operational: a binary, drift-robust watchdog for transduction integrity (a lock that announces its own miscalibration), and a controlled classical emulator of threshold sensing for developing analysis methodology.

Scope. Within that scope, the results are quantitative: the parameter-free threshold location, the critical-fluctuation law, and the long-lived coherent limit cycle are all reproduced on hardware with a per-reading SNR of only ≈ 5 . The identifiability analysis [Eq. (6)] delimits honestly what a threshold shift can assert about the physical system, and the measurement programme of Sec. 5 – in particular the slope-ratio collapse – would turn the single observed threshold into a tested functional relation. For the parent programme on autonomous mesoscopic sensors, where the nonlinearity is physical, the estimators and statistical tools exercised here (threshold bracketing, block-coherent amplitudes, two-time statistics of limit cycles) transfer directly.

7 Conclusion

We have operated a digitally frequency-locked NV magnetometer beyond the flip bifurcation of its own feedback loop, converting the lock into an autonomous period-2 oscillator, and characterized the transition quantitatively: onset at the parameter-free prediction $G_c = 2$ for a self-calibrated loop, sub-threshold fluctuations following the predicted divergence, and a 27-minute coherent limit cycle established by two-time statistics. We derived the measurable content of the threshold, $G_c = 2D_{\text{cal}}/D_{\text{true}}$, together with its exact translation invariance and its non-identifiability between linewidth and amplitude perturbations. The study establishes a low-cost, fully characterized classical testbed – and states its limits – for the broader programme of using deviations from autonomous behaviour as sensing signals in devices where the nonlinearity is physical.

Acknowledgements

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